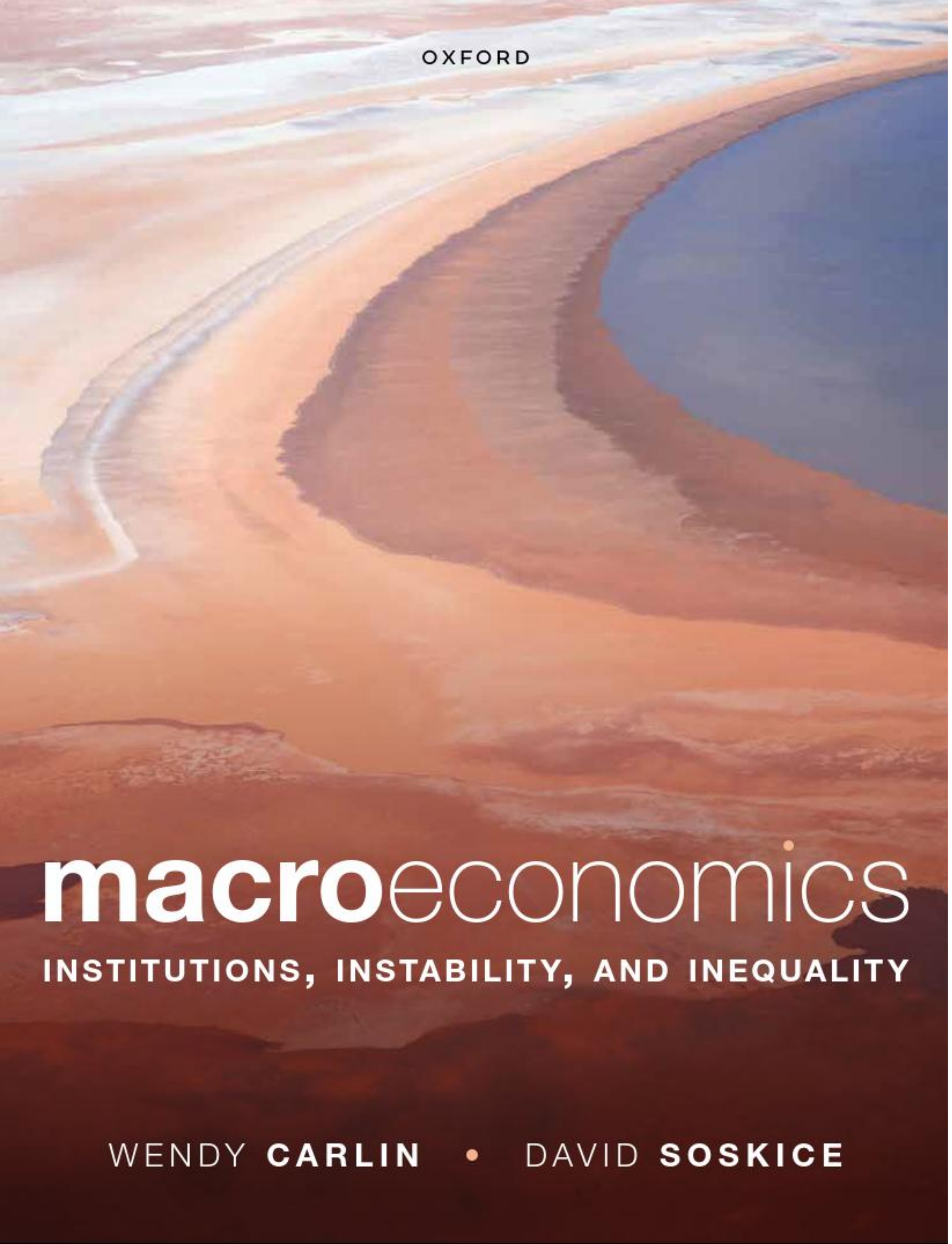

Macroeconomics for International Commerce

PART 1. Flipped Classroom – Week 1 Pre-Class Video Slides

Chapter 2: The Supply Side

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How to use this video

Before class:

- Watch once straight through
- Try to catch the intuition, stop and think
- Bring one question to class
- In class: we will talk about more technical side and applications

Purpose of this video: learn the intuition only. Technical diagrams and applications will be done in class.

What we cover today

- Why unemployment matters;
- Wage-setting intuition;
- Price-setting intuition;
- Equilibrium unemployment/NAIRU;
- The bargaining gap;
- The basic Phillips curve intuition;
- One simple example of positive or negative demand shock.

Question for today
Why can a strong economy sometimes create inflation pressure?

Key mechanism
Unemployment → wage bargaining → firms' prices → inflation

Why unemployment matters

- Unemployment is not only a social problem; it is also a macroeconomic signal.
- When unemployment is low, workers have stronger bargaining power.
- When unemployment is high, workers fear job loss and wage pressure weakens.

Low unemployment
→ stronger worker bargaining
→ higher wage pressure

High unemployment
→ weaker worker bargaining
→ lower wage pressure

Lower unemployment usually strengthens workers' bargaining position. Higher unemployment weakens it.

Wage-setting (WS) intuition

- Firms cannot perfectly monitor workers, so wages must also motivate effort.

High unemployment

Weak bargaining position

A worker who fears long unemployment is less able to demand high wage growth.

cost of job loss

Lower ⟷ Higher

Low unemployment

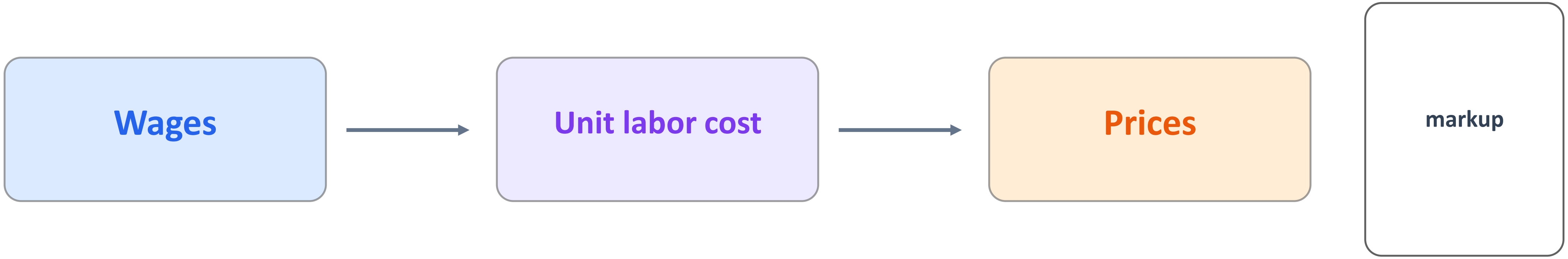
Strong bargaining position

When jobs are easy to find, workers can push for higher wage growth.

Wage-setting curve = wage needed to recruit and motivate workers at each unemployment level.

Price-setting (PS) intuition

- Firms set prices as a markup over costs. With labor as the main input, wages matter for prices.
- Firms set prices with costs, productivity, and a profit margin in mind.



Simple rule: if wages grow faster than productivity, firms face higher labor costs and tend to raise prices.

Equilibrium unemployment: WS = PS

- Equilibrium unemployment is where workers' wage claims and firms' price-setting are consistent.

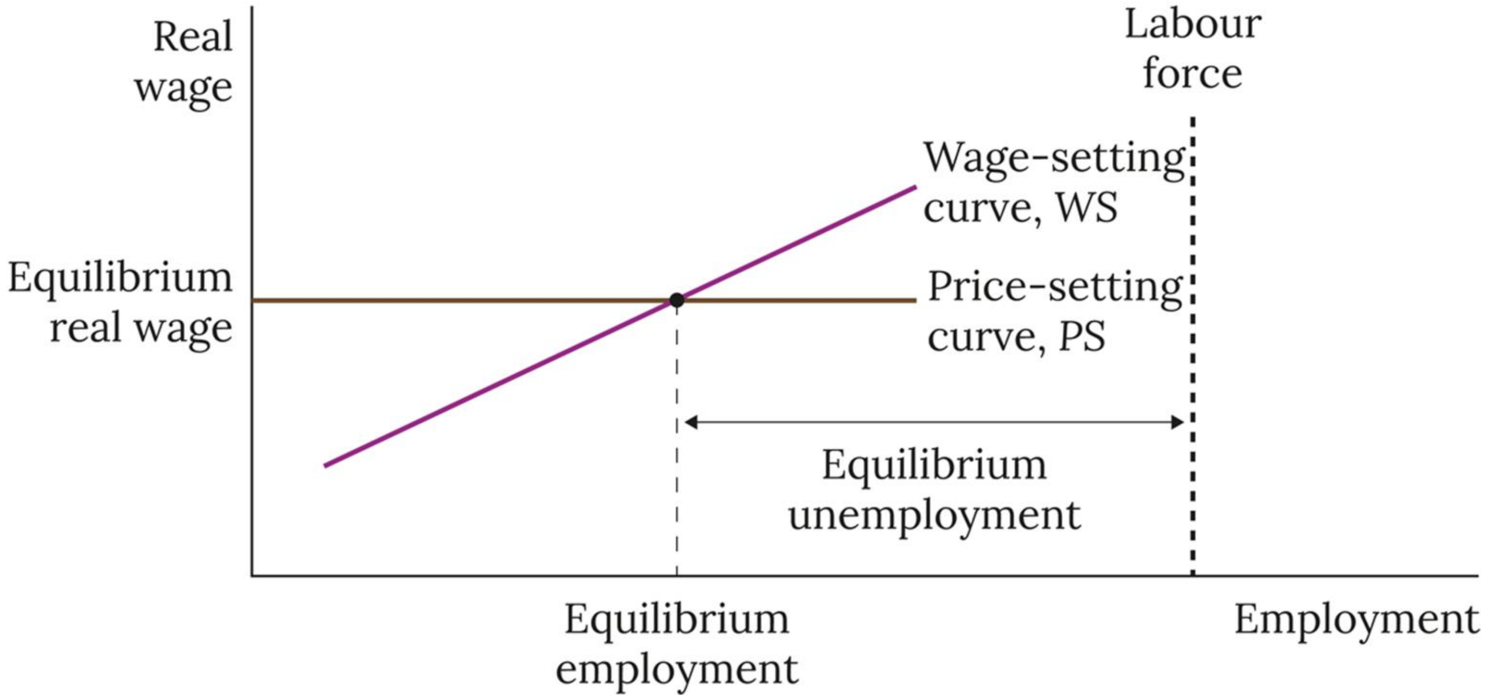


Figure 2.6 Supply-side equilibrium: the intersection of the wage-setting and price-setting curves.

NAIRU: the inflation-stabilizing unemployment rate

- NAIRU means the unemployment rate at which inflation is not accelerating.

Unemployment below NAIRU	**Unemployment at NAIRU**	**Unemployment above NAIRU**
Workers stronger Wage pressure rises Inflation tends to increase	Wage claims and price setting are consistent Inflation remains stable	Workers weaker Wage pressure falls Inflation tends to decrease

Important: NAIRU is a model concept, not directly observed like the actual unemployment rate.

The bargaining gap

- The bargaining gap appears when the real wage workers try to obtain differs from the real wage firms are willing to pay.

Below NAIRU	At NAIRU	Above NAIRU
Unemployment is low. Workers have stronger bargaining power. Wage pressure rises.	Wage-setting and price-setting are consistent. Inflation pressure is stable.	Unemployment is high. Workers have weaker bargaining power. Wage pressure falls.
positive bargaining gap	**no bargaining gap**	**negative bargaining gap**

Inflation pressure appears when the economy is away from the WS–PS equilibrium.

Basic Phillips Curve intuition

- The Phillips Curve summarizes how inflation changes when output/employment is away from equilibrium.

current inflation = last period's inflation + inflation pressure

Positive gap

Output above equilibrium → unemployment low → inflation rises.

No gap

Output at equilibrium → inflation remains stable.

Negative gap

Output below equilibrium → unemployment high → inflation falls.

In words: too much demand creates inflation pressure; too little demand reduces it.

Simple example: positive demand shock

- A boom can lower unemployment below equilibrium and create rising inflation pressure.

Initial state → Demand rises → Unemployment falls → Inflation ↑

Step	Unemployment	Wage pressure	Inflation
Equilibrium	6%	normal	2%
After boom	3%	stronger	4%
If gap persists	below NAIRU	still strong	rises again

Class question: What policy response could stop inflation from continuing to rise?

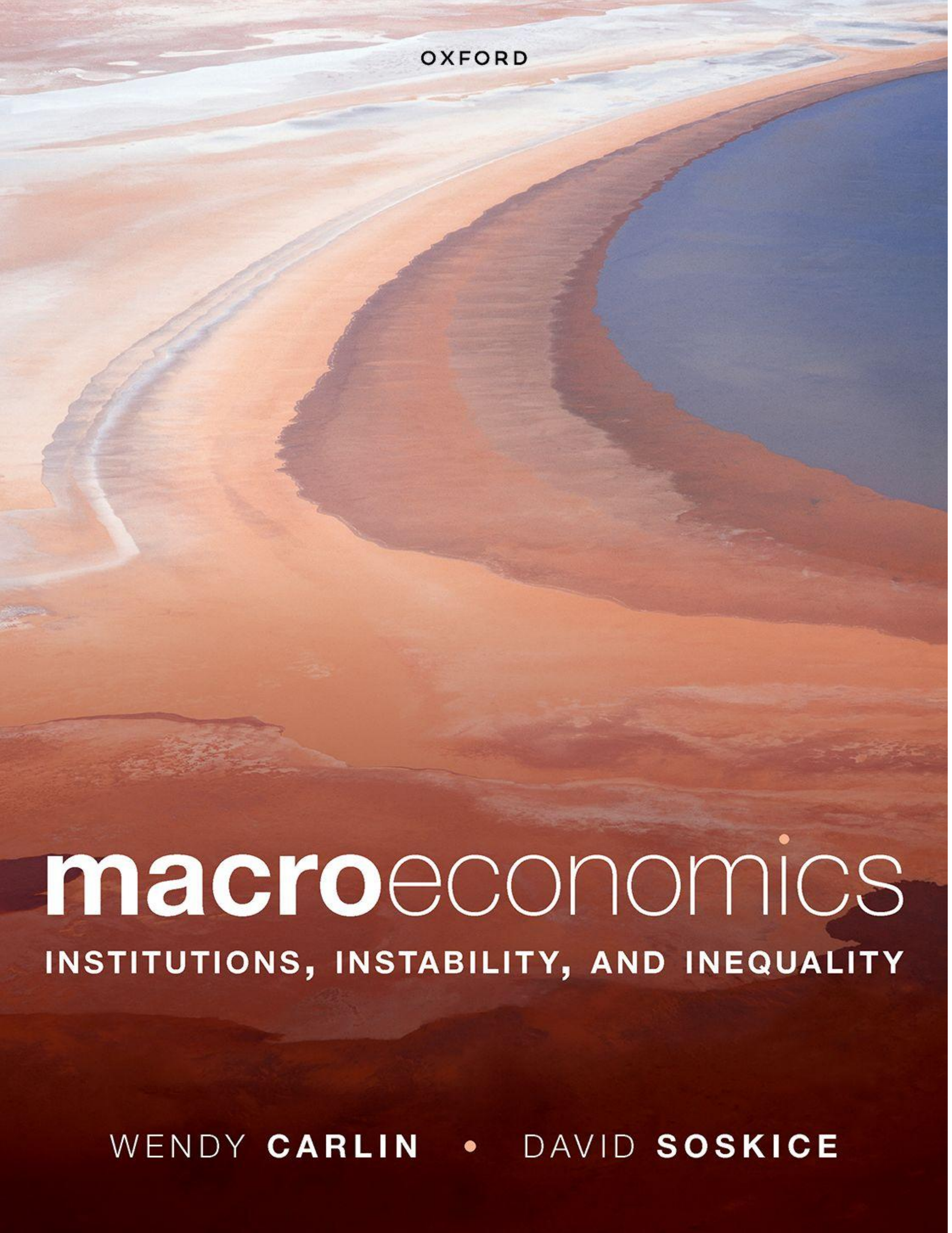
Readiness check

After watching, you should be able to answer these before class.

- Why does low unemployment increase workers' bargaining power?
- How can faster wage growth translate into higher prices?
- What does equilibrium unemployment / NAIRU mean?
- What is the bargaining gap in plain language?
- In a positive demand shock, why can inflation rise?

Pre-class takeaway: **you do not need the technical derivation yet; we will add the diagrams and formal model in class.**

For class, remember the causal chain: **unemployment → wage bargaining → prices → inflation.**



Macroeconomics for International Commerce

PART 2. Full In-Class Slides

Chapter 2: The Supply Side

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Objectives

By the end of this chapter, students should understand the following:

- The WS-PS Unemployment model
- The Phillips Curve and its properties
- How medium-run unemployment is determined
- How inflation behaves if output is away from equilibrium
- The effects of demand and supply shocks in the absence of a policy maker
- How shifts in WS-PS model impact inequality and are shown in Lorenz curve

Overview: Unemployment of OECD Countries

Unemployment varies across countries and time

- There is a large amount of heterogeneity in unemployment across the high-income countries when they faced the same economic event

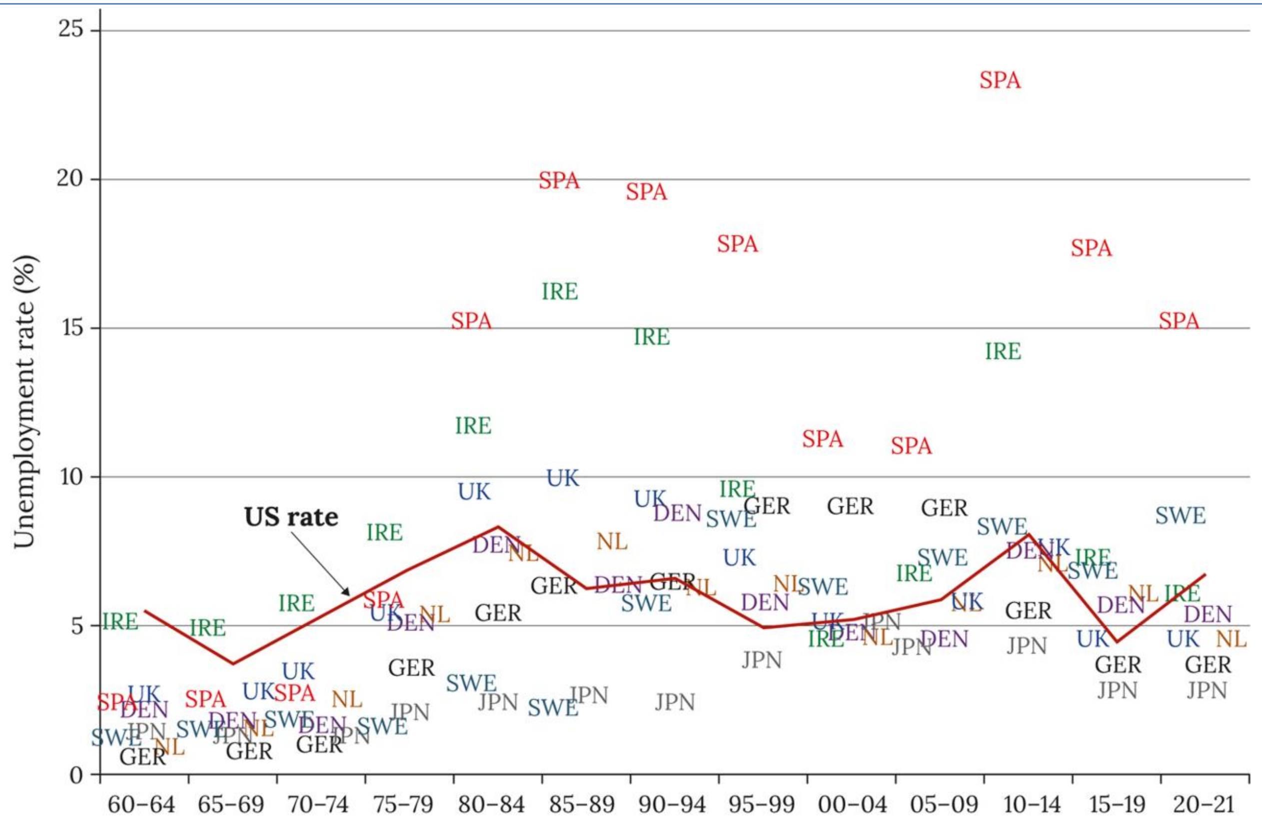


Figure 2.1 Trends and heterogeneity in unemployment for selected OECD economies, 1960–2021. *Source*: Howell et al. (2007), Figure 1.1, used until 1995. Updated to 2021 using OECD harmonized unemployment rates (data accessed June 2022).

Overview: Labour market

With involuntary unemployment, the labour market does not clear

- There are workers who are willing to work at the going wage but cannot find a job

Firms considers two aspects when hiring a worker:

- Recruitment problem
  - Reservation wage: the minimum wage the employer can pay to recruit a worker
- Motivation problem
  - Firms want to ensure workers generate enough effort, but they cannot perfectly monitor workers (conflict of interest between employers and workers)
  - No-shirking wage: the wage that needs to be paid to motivate workers to exert effort; any wage higher than the reservation wage
- Outcome is involuntary unemployment because employer pays a wage above the reservation wage to ensure worker supplies the required effort

Overview: Wage-Setting Curve

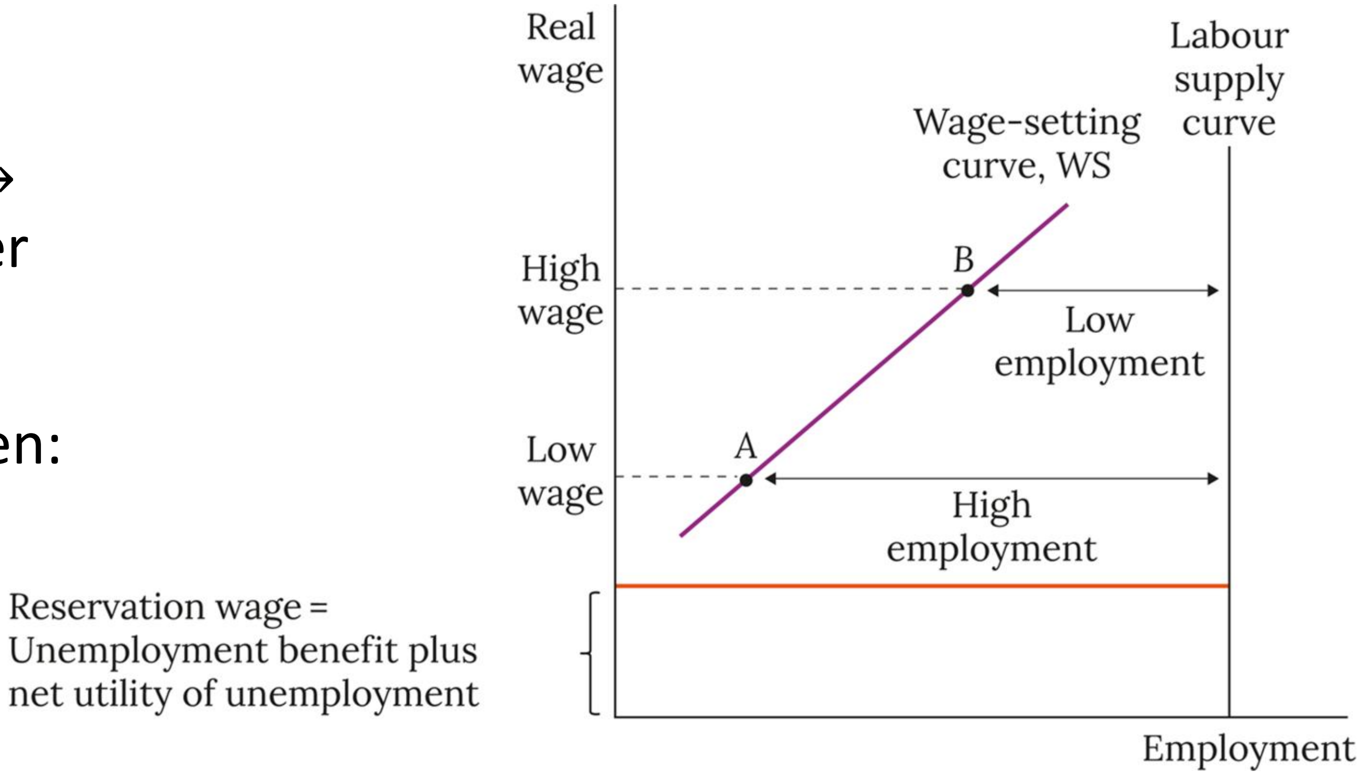
The wage-setting curve is upward sloping and above the reservation wage

- 'A': High U → high cost of job loss → only low w is required
- 'B': Low U → low cost of job loss → high w is required to induce worker effort

The wage-setting curve shifts up when:

- Unemployment benefit increases

Figure 2.2 The wage-setting curve, *WS*.



Overview: Labour Supply Curve

The labour supply curve is vertical (assumption)

1. balanced income and substitution effect

2. when wage increases, the tendency of workers to enter the labour force is offset by a fall in hours of work of those who are employed

Involuntary unemployment: the horizontal gap between WS curve and LS curve

Empirical wage setting curve (US) highlights convexity of WS:

- As unemployment falls toward zero, wage rises by more

- Always involuntary unemployment, so WS is asymptotic to the labour supply curve.

- Linear WS is a simplification (Ch. 15).

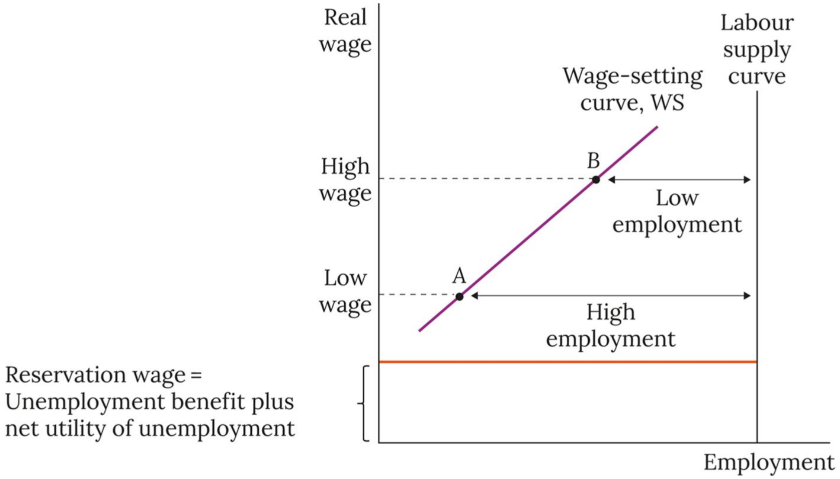


Figure 2.2 The wage-setting curve, *WS*.

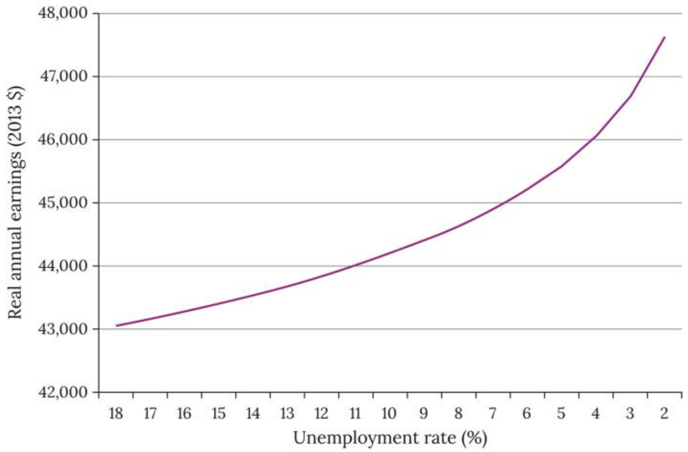


Figure 2.3 A wage-setting curve estimated for the US economy (1979–2013).
Note: The unemployment rate is measured from high on the left to low on the right.
Source: Estimated by Stephen Machin (UCL 2015) from Current Population Survey microdata from the Outgoing Rotation Groups for 1979 to 2013.

Overview: Union Wage Setting

There is heterogeneity in union power across countries.

- High union power → exercise bargaining power → higher WS curve → high unemployment
- High union power → exercise bargaining restraint → lower WS curve → low unemployment

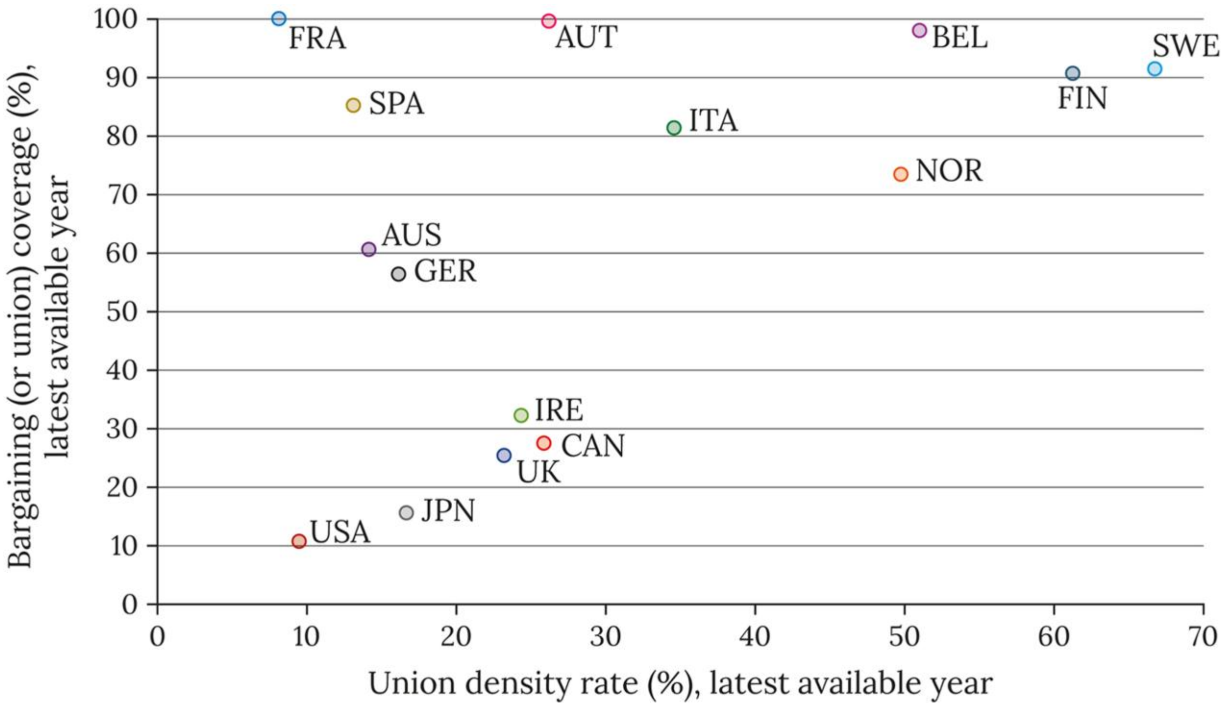


Figure 2.4 The importance of unions in wage setting: selected OECD countries.
Source: OECD Statistics (data accessed July 2020).
AUS Australia, AUT Austria, BEL Belgium, CAN Canada, DEU Germany, ESP Spain, FIN Finland, FRA France, IRL Ireland, ITA Italy, JPN Japan, NOR Norway, SWE Sweden, UK United Kingdom, US United States. See footnote 9.

Overview: Price-Setting Curve

Supply of Labour is reflected by Wage-Setting (WS) Curve

Demand for Labour is reflected by Price-Setting (PS) Curve

- Flat PS: we assume constant marginal productivity of labour

Additional assumptions:

- Labour is the only input
- Firms require a fixed profit margin to find it profitable to employ workers

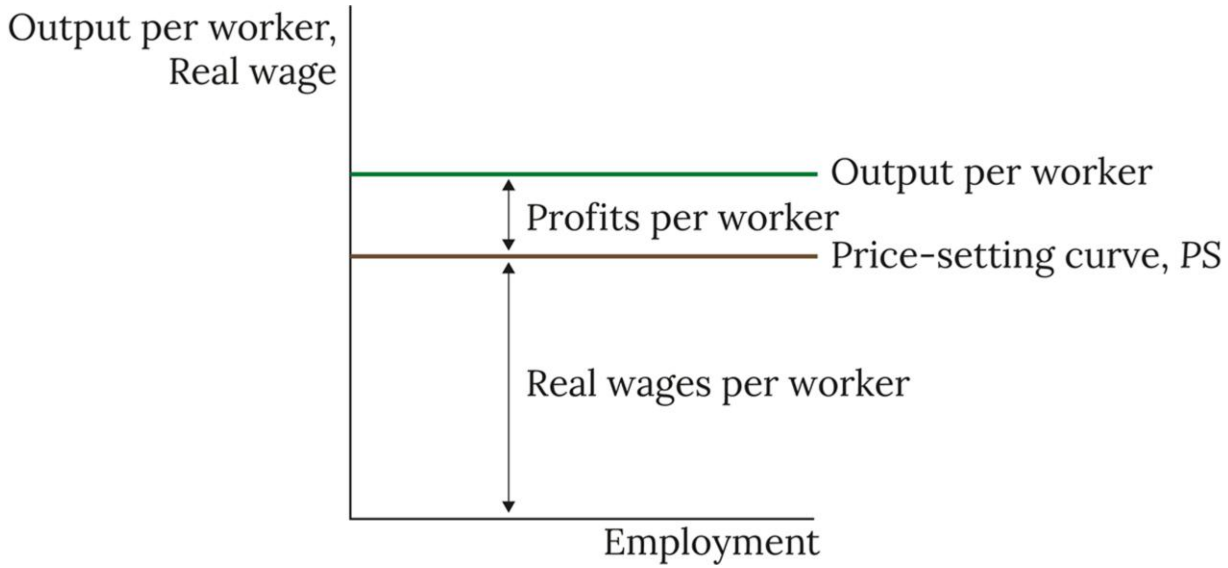


Figure 2.5 The price-setting curve, *PS*.

Overview: WS-PS Model

Labour Market Equilibrium: $WS = PS$

- Equilibrium unemployment rate $\equiv \dfrac{\text{equilibrium unemployment}}{\text{labour force}}$

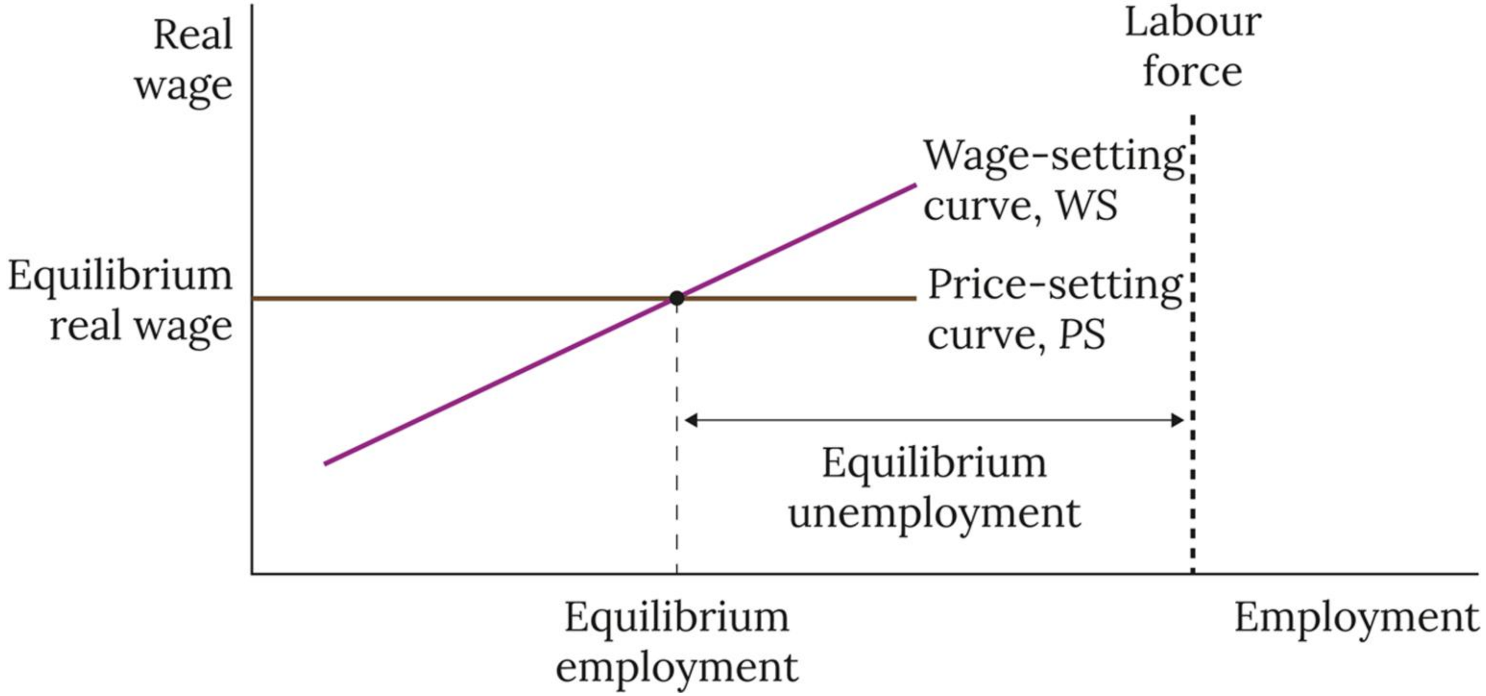


Figure 2.6 Supply-side equilibrium: the intersection of the wage-setting and price-setting curves.

Overview: Shifts in PS and WS Curves

Equilibrium w and U change due to shifts in the WS and PS

- $\uparrow$ unemployment benefit $\rightarrow WS$ shifts up $\rightarrow \uparrow U$, but w is fixed by PS
- $\uparrow$ Mark-up $\rightarrow PS$ shifts down $\rightarrow \uparrow U, \downarrow w$

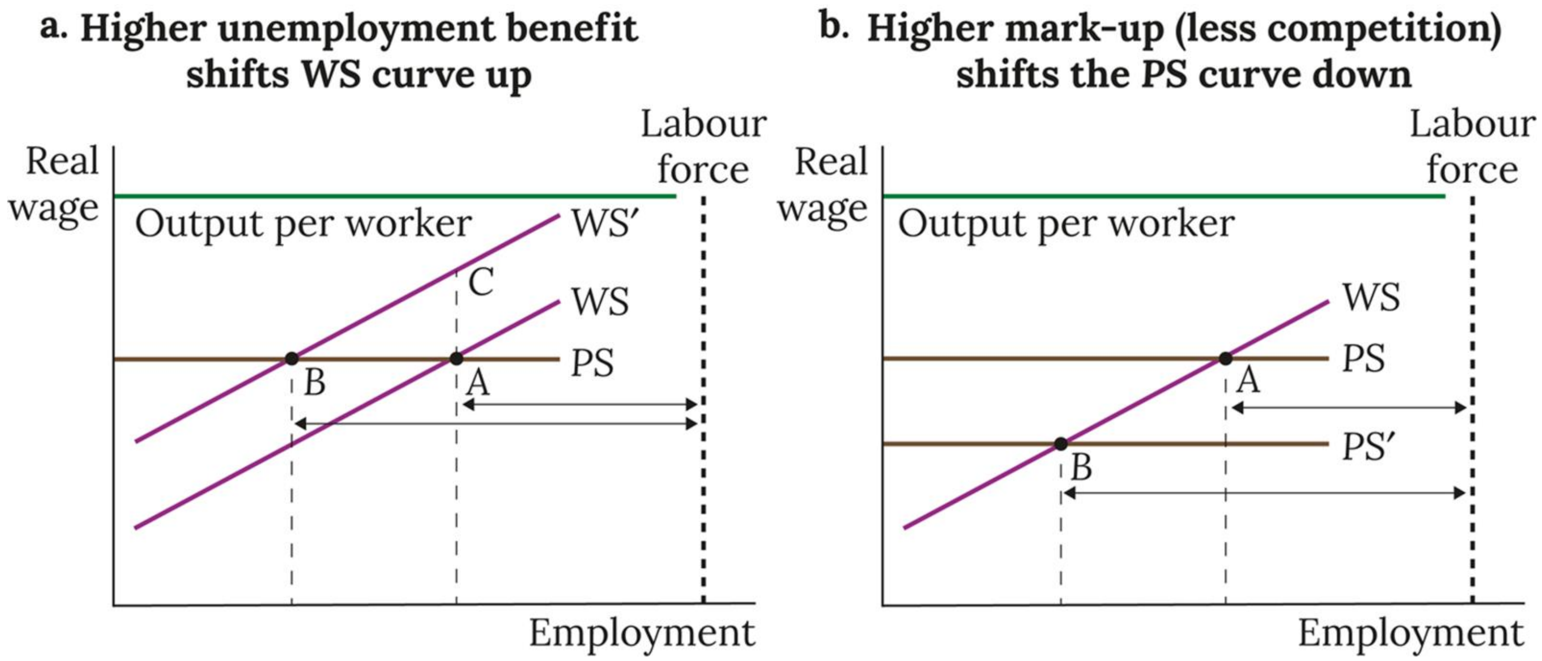


Figure 2.7 The impact on equilibrium unemployment of shifts in the *WS* and *PS* curves. Visit www.oup.com/he/carlin-soskice to explore the effect of shifting the *WS* and *PS* curves on unemployment with Animated Analytical Diagram 2.7.
Note: the double-headed arrows show equilibrium unemployment for each case.

Overview: Wage and Price Rigidity

Wage and price rigidity: nominal wages and prices are adjusted infrequently

Demand Shock under fixed W & P: Output and employment move away from medium run equilibrium → Role for the policy maker!

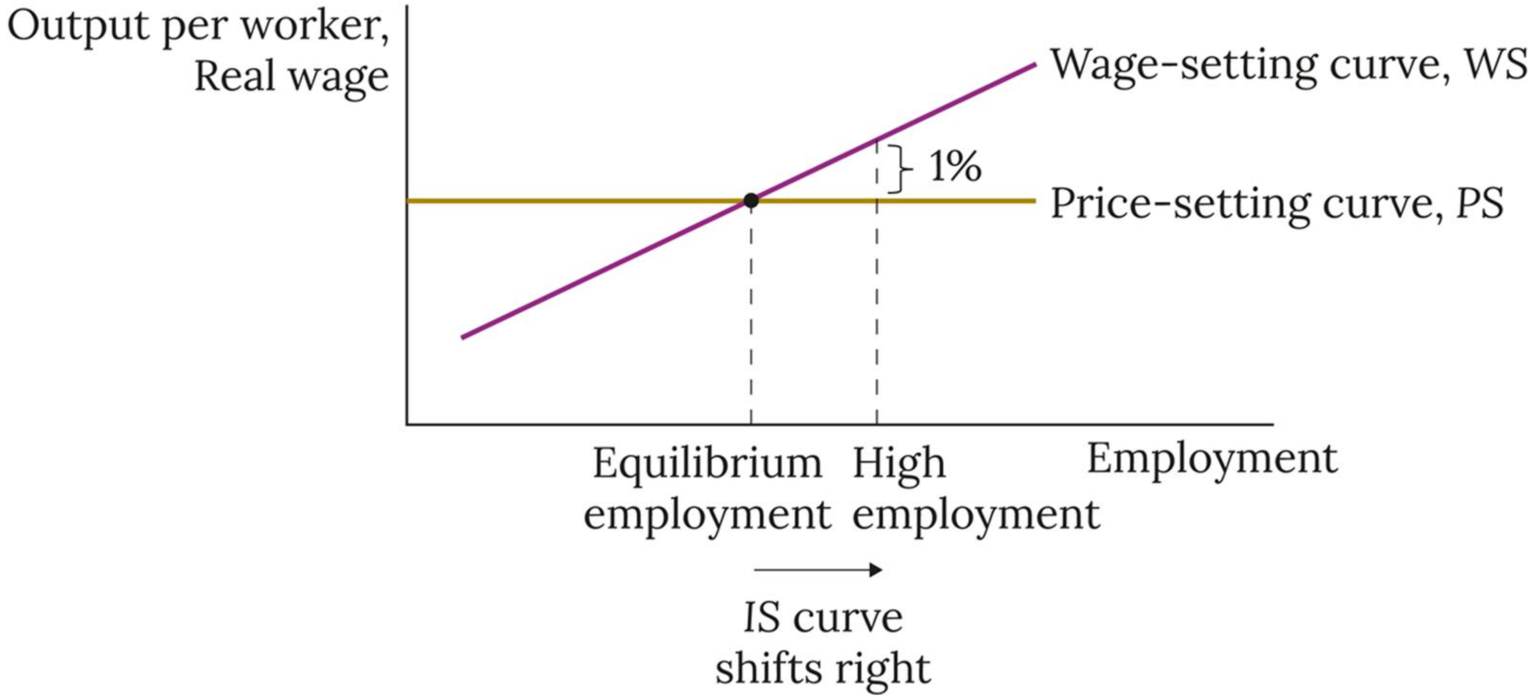


Figure 2.8 The response of wages and prices to a cyclical upswing.

Overview: Actual and Equilibrium Unemployment

Actual unemployment and Equilibrium Unemployment (NAIRU):

- Differences in experience between Europe and the UK, US
- The NAIRU is much less volatile than actual unemployment

Note: The non-accelerating inflation rate of unemployment (NAIRU) is a theoretical level of unemployment below which inflation would be expected to rise.

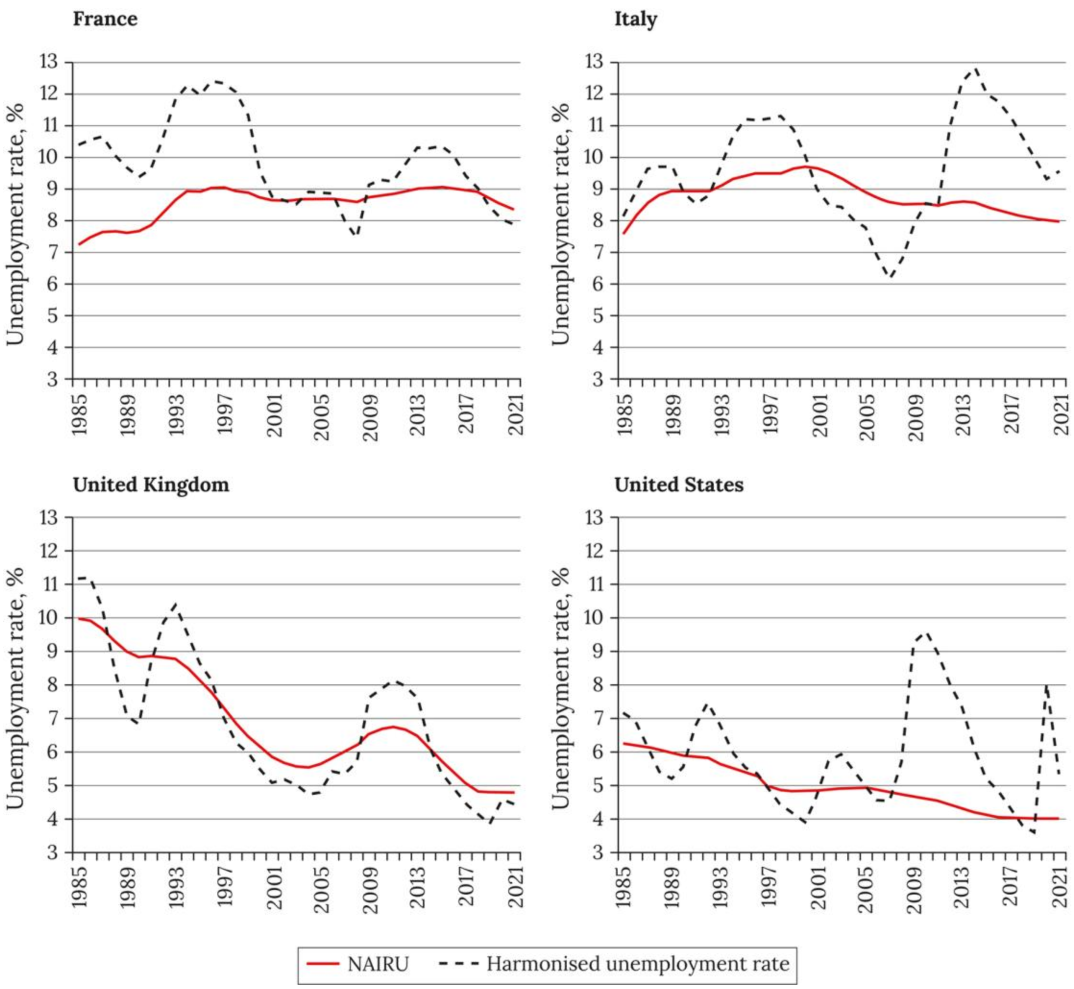


Figure 2.9 Non-Accelerating Inflation Rate of Unemployment (NAIRU) and harmonized unemployment rates in France, Italy, the United Kingdom, and the United States: 1983–2021. *Source*: OECD Economic Outlook (May 2021 edition).

Overview: Inflation

Unemployment and Inflation: Decline in Inflation from the 1970s to mid 2000s in France and Italy is consistent the trends of unemployment as displayed in Figure 2.9

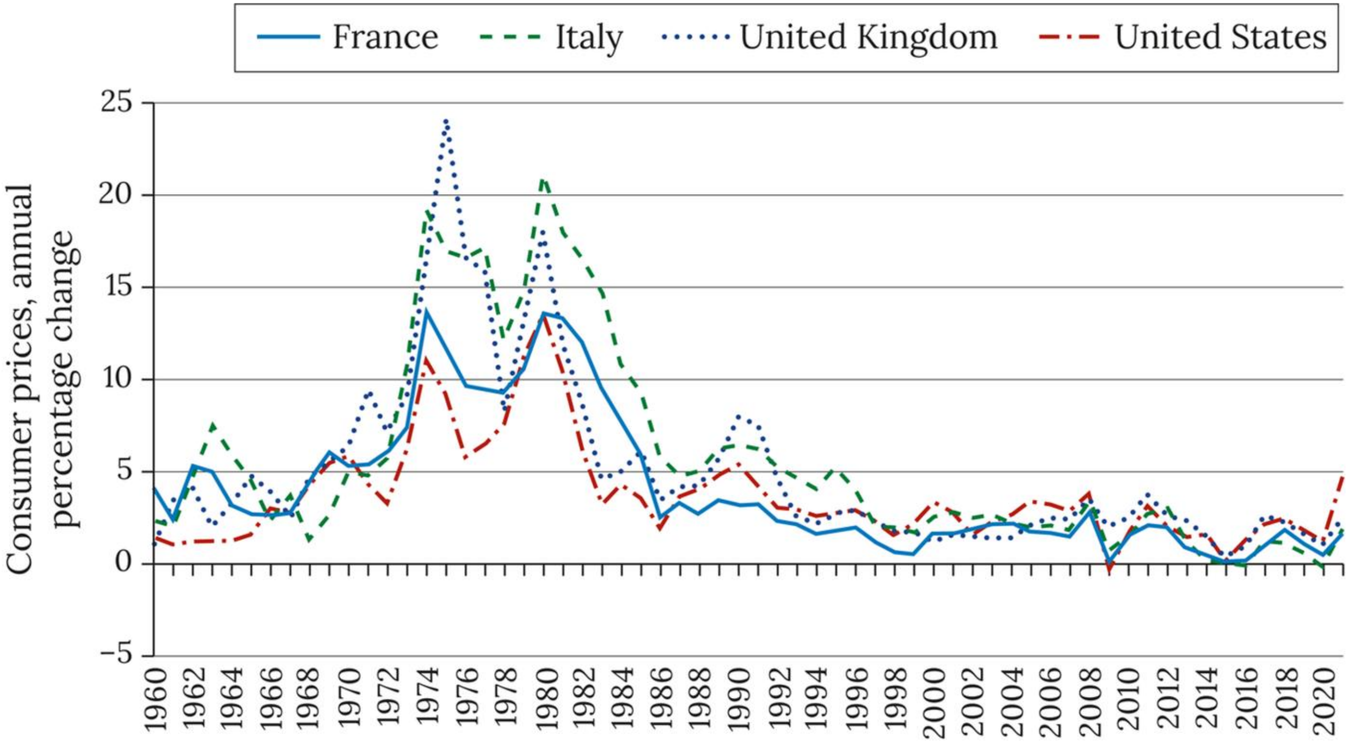


Figure 2.10 Consumer price inflation rates in France, Italy, the United Kingdom, and the United States: 1960–2021.
Source: OECD Monthly Economic Indicators (accessed July 2022).

Modelling: Wage-Setting Curve (Math)

WS real wage equation:

$$w^{WS} = \frac{W}{P^E} = B(N, \boldsymbol{z}_w)$$

- P^E: the expected price level
- N: level of employment
- $\boldsymbol{z}_w$: a set of wage-push variables

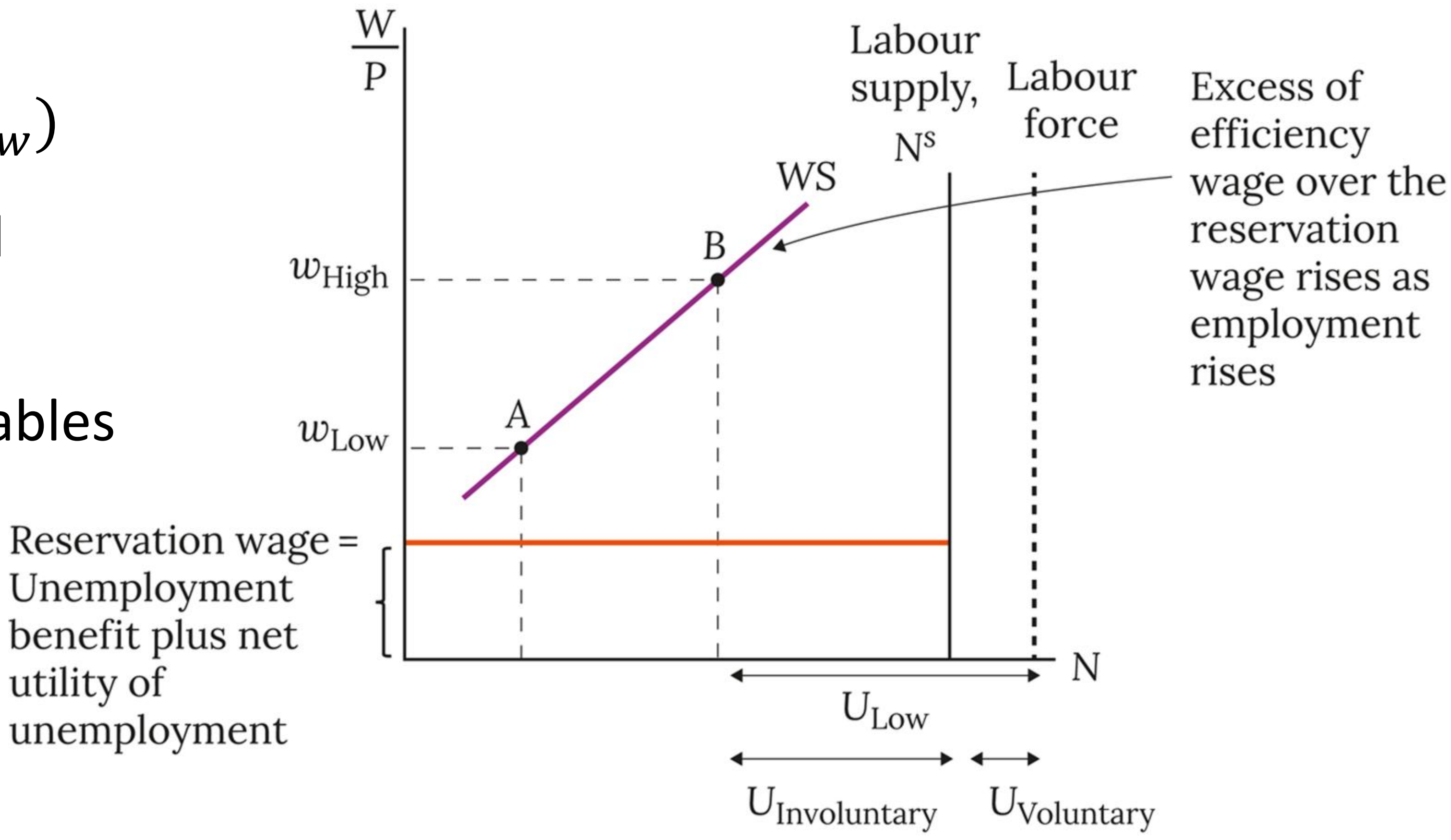


Figure 2.11 The *WS* curve, the reservation wage, voluntary and involuntary unemployment. Visit www.oup.com/he/carlin-soskice to explore the WS curve, the reservation wage, voluntary and involuntary unemployment with Animated Analytical Diagram 2.11.

Modelling: Shifts in WS Curve

Wage-push factors ($\mathbf{z}_w$) shifts the WS Curve up when:

- Efficiency wage factors:
  - Increase in Unemployment benefits or its duration or a fall in cost of applying for it or in the eligibility criteria
  - Increase in the net utility of unemployment
  - Increase in the cost of effort
  - Monitoring of work effort or firing a worker who shirks becomes more difficult and costly
- Union-related:
  - Increase in legal protection for unions
  - Stronger union power, covering higher proportion of employees
  - Unions cease exercising bargaining restraint

Modelling: Price-setting curve

When a firm faces a downward-sloping demand curve, they set a price to maximize their profits:

Price-setting formula:

- $P = \left(1 + \frac{1}{\eta - 1}\right)\left(\frac{W}{MPL}\right) = (1 + \mu^C)\left(\frac{W}{MPL}\right)$
- η: Elasticity of demand
- μ^C: Mark-up on MC

Rearrange in terms of real wage, we get the price-setting real wage formula:

- $\frac{W}{P} = MPL(1 - \text{markup on } P) = MPL\ (1 - \mu)$,

where $(1 - \mu) = \frac{1}{1+\mu^C}$

Modelling: Downward sloping PS Curve

Downward sloping price-setting real wage curve is a fraction of MPL

But here we simplify the model and apply the horizontal PS curve

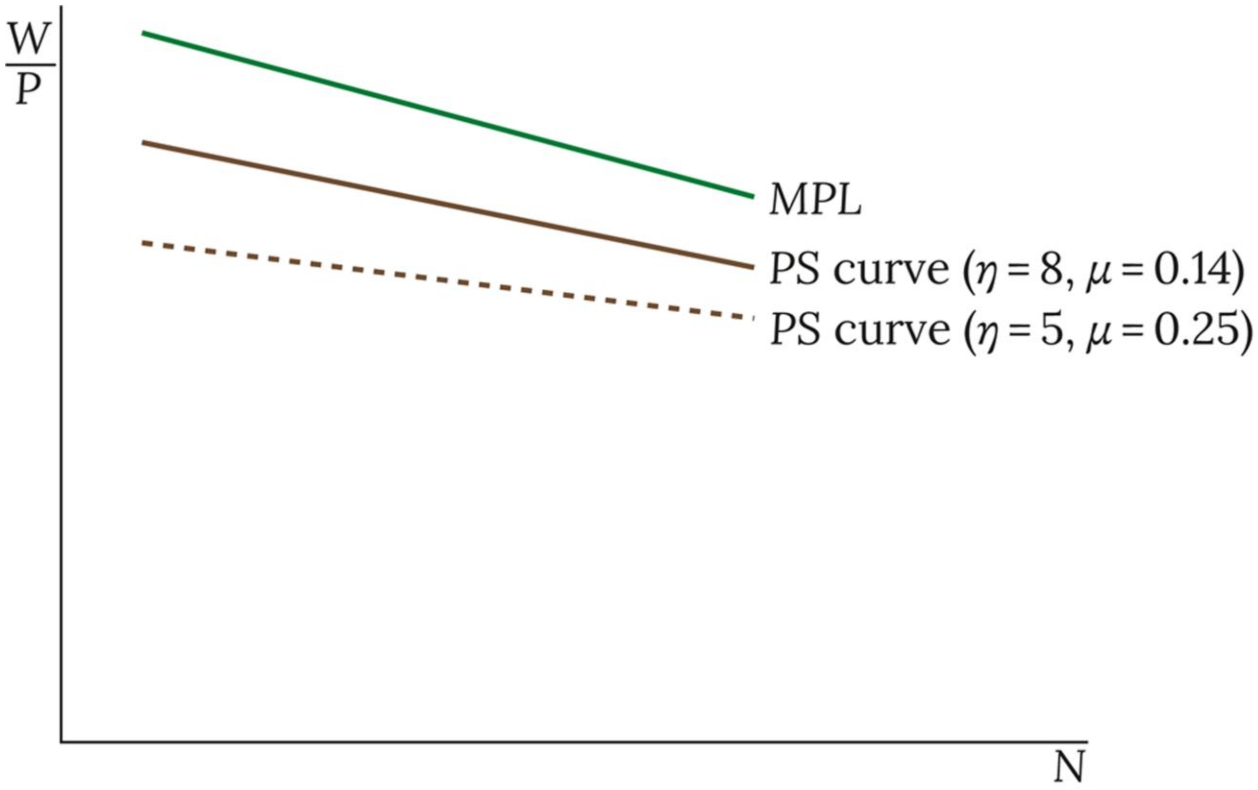


Figure 2.12 Relationship between the *MPL*, the price elasticity of demand (η), and the *PS* curve.

Modelling: Horizontal PS Curve

To derive the horizontal PS Curve, we assume constant productivity and mark-up

Hence, Price-setting formula can then be written as:

- $P = (1 + \mu^C)\left(\frac{W}{\lambda}\right)$
- $\lambda = \frac{y}{N}$, output per worker

Rearranging, we can write PS real wage equation in terms of mark-up on price:

- $w^{PS} = \frac{W}{P} = \lambda(1 - \mu)$,

where $(1 - \mu) = \frac{1}{1+\mu^C}$

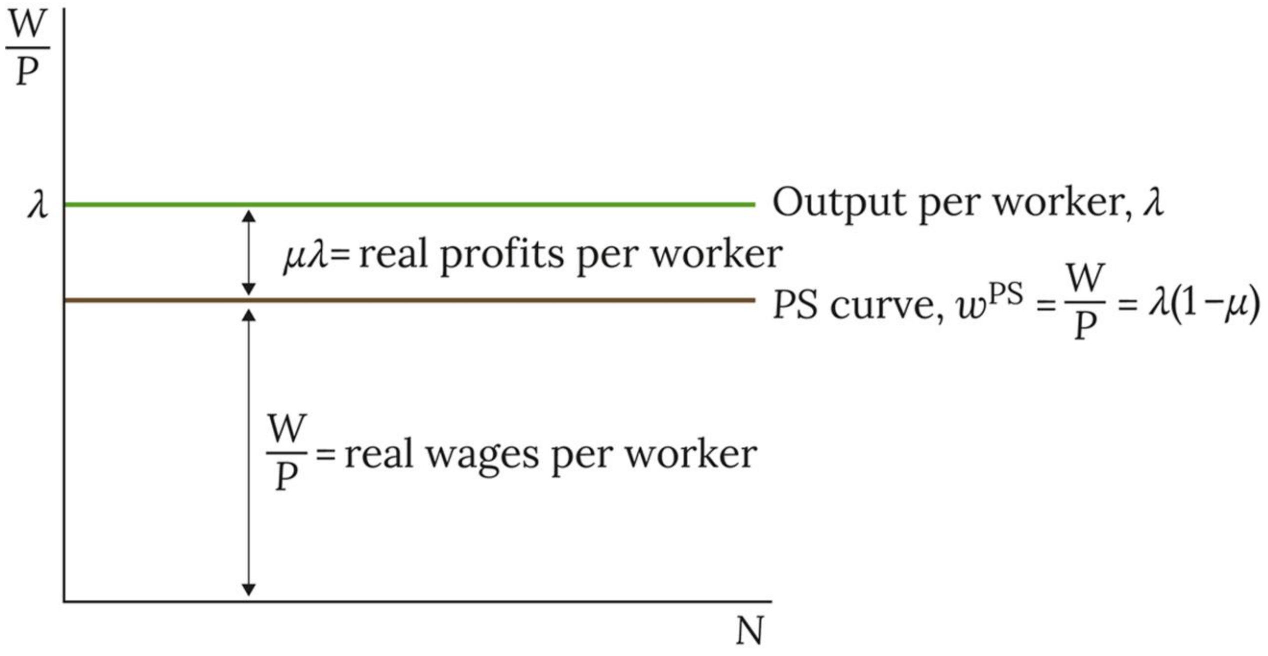


Figure 2.13 The price-setting real wage curve: *PS*.

Modelling: Shifts in PS Curve

Another way to write PS Curve is:

- $w^{PS} = \lambda F\left(\mu, \boldsymbol{z}_p\right)$
- where $\boldsymbol{z}_p$ is a set of price push variables

The PS curve shifts down, reducing equilibrium unemployment, when there is:

- A rise in the tax or oil price wedge (real consumption wage – real product wage)
- A rise in the markup (μ), e.g., due to weaker competition policy rules or enforcement
- A fall in productivity (λ)

Modelling: Labour Market Equilibrium

Labour Market Equilibrium:

- $w^{WS} = w^{PS}$
- $B(N, \boldsymbol{z}_w) = \lambda F(\mu, \boldsymbol{z}_p)$

Equilibrium unemployment arises from the structural or supply-side features of the economy

- Supply-side policies or structural changes can therefore impact such equilibrium rate

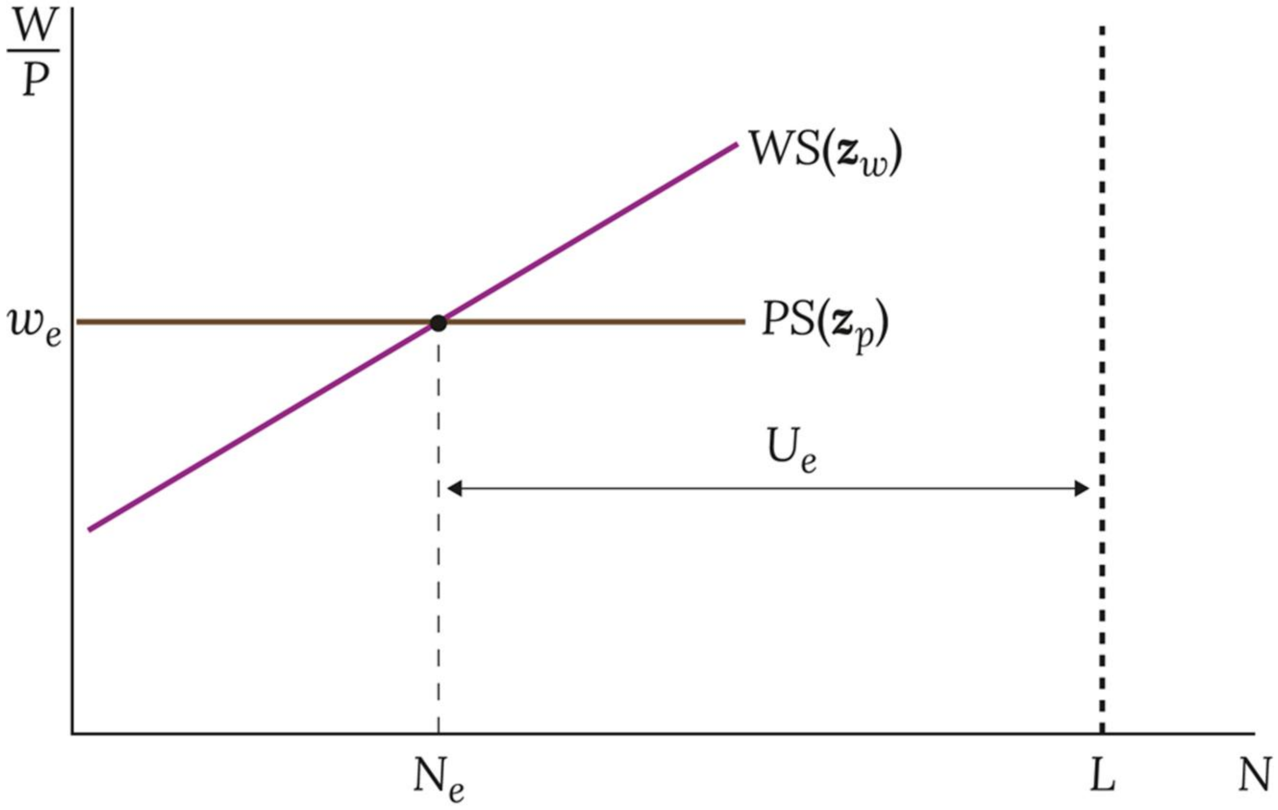


Figure 2.14 Equilibrium employment and unemployment: N_e and U_e.

Modelling: Supply-side equilibrium and Inequality

- Gini coefficient measures inequality as differences in income between people: one-half the average of the differences divided by the average income.
- Example of 3-person economy: Gini = $0.5\left(\frac{6.67}{6}\right)$=0.56
- Equal to 1 if a single person owns everything; 0 if all have the same income
- Lorenz curve is visualization of Gini coefficient in which people are ordered along the horizontal axis by their income; vertical shows cumulative share of income; 45° line shows equal incomes
- 2 determinants of inequality can be mapped from WS-PS model to Lorenz curve and Gini – unemployment and markup

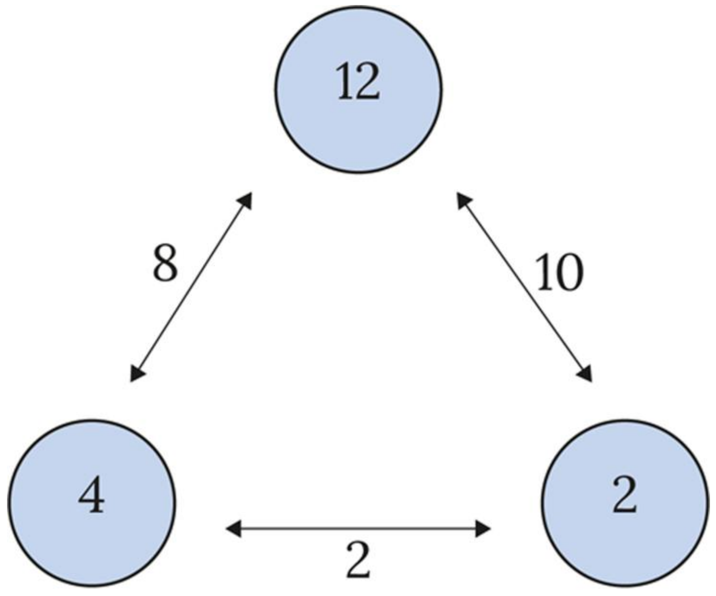


Figure 2.15 Income differences among pairs of households.

Modelling: Supply-side equilibrium and Inequality

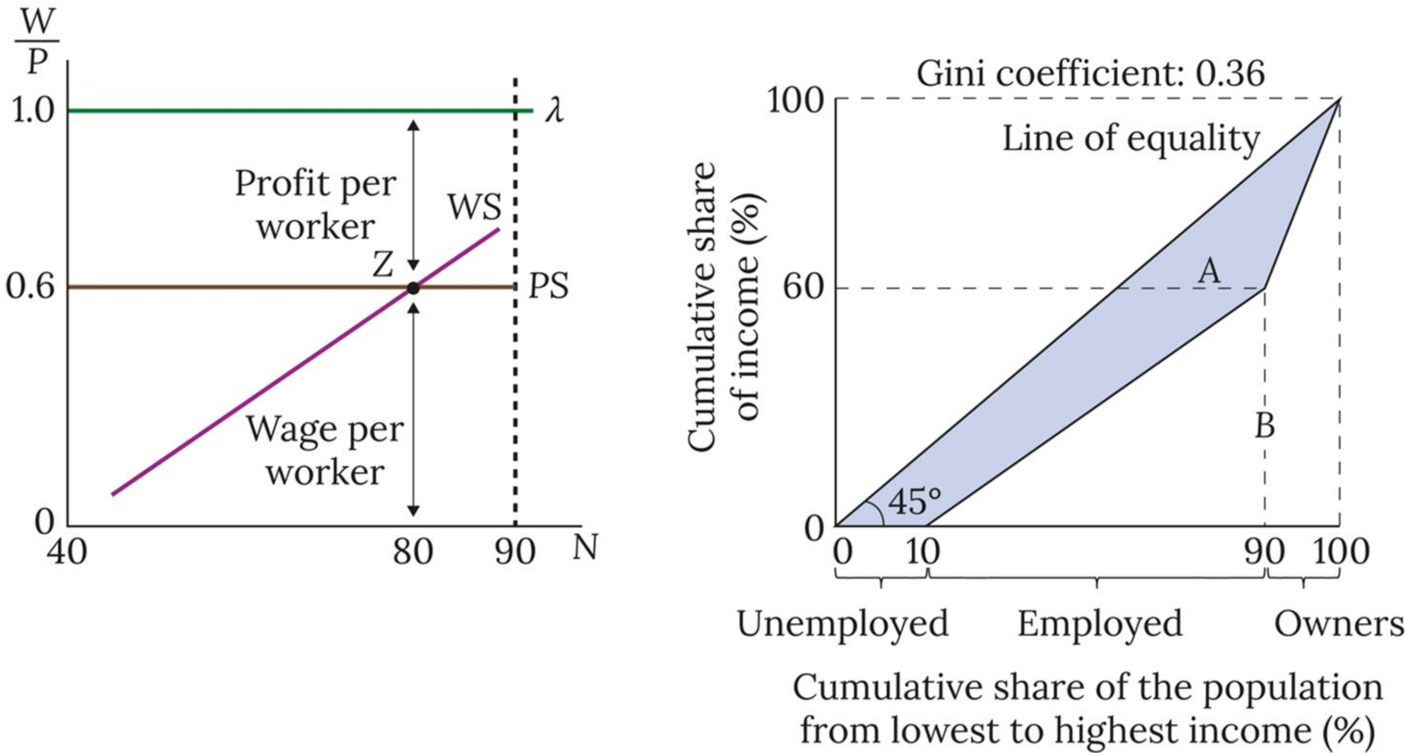


Figure 2.16 The distribution of income at labour and product market equilibrium. Visit www.oup.com/he/carlin-soskice to explore distribution of income at labour and product market equilibrium with Animated Analytical Diagram 2.16.

Map WS-PS model to the Lorenz curve and Gini coefficient

Assume there are 3 groups of income recipients in the economy:

- 10 unemployed
- 80 employed
- 10 owners of firms

Inequality widens, if there is:

- A rise in the mark-up (μ)
- A structural change that raise equilibrium unemployment

Why we need the PC

- WS–PS: intersection gives equilibrium unemployment (the NAIRU)
  - *But what happens if unemployment is not at equilibrium?*
  - *How do wages and prices react?*
- **Phillips Curve (PC)**

Note: The non-accelerating inflation rate of unemployment (NAIRU) is a theoretical level of unemployment below which inflation would be expected to rise.

Intuition

Step A – From WS–PS to wage pressure

- Low unemployment → workers feel secure → demand higher wages.
- High unemployment → workers fear job loss → accept lower wage growth.
- So, wage growth depends on the *output gap* (employment vs. equilibrium).

Intuition

Step B – From wages to prices

- Firms set prices as a markup on wages.
- If wages grow faster, firms raise prices → inflation rises.
- If wages grow slowly, inflation falls.

Intuition

Step C – Put together

- Inflation this period = last period's inflation (because of expectations) + "bargaining gap" (output gap).

- That's the Phillips Curve:

$$\pi_t = \pi_{t-1} + \alpha(y_t - y_e)$$

The bargaining gap is the difference between the real wage workers want (based on their bargaining power at a given unemployment level) and the real wage firms are willing to pay (based on the price-setting curve).

In the Phillips Curve, it is equivalent to the output gap: how far actual output/employment is from the equilibrium level.

Real-life analogy

Concert tickets example:

- Imagine a small concert hall with 100 seats.
- If 100 people show up (equilibrium), ticket prices stay steady.
- If 150 people show up (excess demand), organizers raise prices next time.
- If only 50 people come (low demand), they cut prices.

That's the Phillips Curve logic: "too much demand" → inflation rises; "too little" → inflation falls.

Modelling: Aggregate Demand Shocks

The economy moves away from the medium-run equilibrium ($w^{WS} = w^{PS}$) due to AD fluctuations

- Demand-driven cycles: output adjusts (instead of prices & wages) due to AD shocks
- Price stickiness: Firms' reluctance to change prices when AD changes

Assume firms only adjust wages at the start of each 'wage round'. Then, an AD shock's effects are modelled as follows:

- AD shock → output and employment change
- Next wage round → nominal wages change
- Immediately after the wage round → prices change

Modelling: Derivation of Phillips Curve (Math)

This behavior is also captured by the Phillips curve

- Output Gap $= y_t - y_e$ (output less equilibrium level)
- Wage setters change wages to cover the previous period's price increase and to reflect the output gap (e.g., due to AD shocks):

$$(\Delta W/W)_t \approx (\Delta P/P)_{t-1} + \alpha(y_t - y_e) \qquad \text{(Wage inflation)}$$

- Firms set prices immediately after wages have been set (productivity (λ)):

$$(\Delta P/P)_t = (\Delta W/W)_t - (\Delta\lambda/\lambda)_t \qquad \text{(Price inflation)}$$

- Substituting the wage inflation into the price inflation , we get the Phillips Curve:

$$(\Delta P/P)_t = (\Delta P/P)_{t-1} + \alpha(y_t - y_e)$$

$$\underbrace{\pi_t}_{\text{current inflation}} = \underbrace{\pi_{t-1}}_{\text{lagged inflation}} + \underbrace{\alpha \underbrace{(y_t - y_e)}_{\text{output gap}}}_{\text{bargaining gap}}$$

Modelling: Derivation of Phillips Curve (Graph)

Positive AD shock → ↑ Employment → ↑ Worker's power in the labour market → Wage-setter sets higher wages (4%) to cover π_{t-1} (2%) and the bargaining gap (2%)

Price setters set higher prices to cover higher wage costs (↑ 4%) → Price Inflation (from 2% up to 4%)

A positive output gap causes Wage and Price Inflation to rise

- Joining points A and B in the π-y space gives us the PC

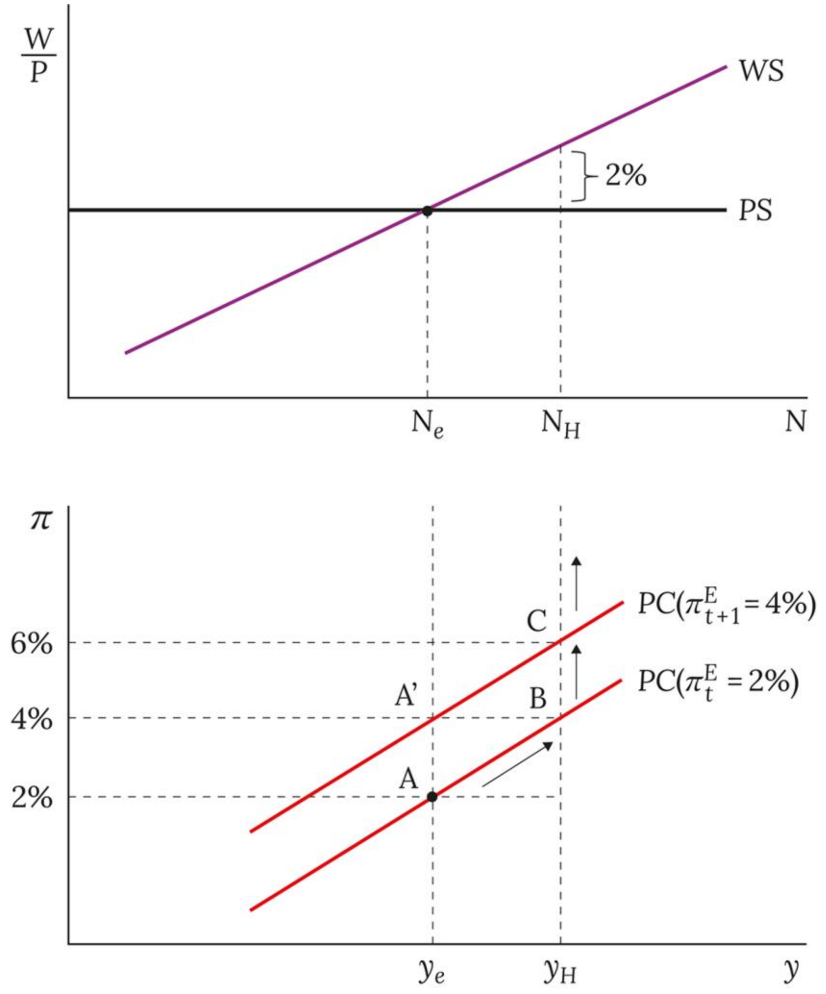


Figure 2.17 The derivation of the Phillips curve. Visit www.oup.com/he/carlin-soskice to explore the derivation of the Phillips curve with Animated Analytical Diagram 2.17.

Modelling: Derivation of Phillips Curve (Graph)

Then, the Phillips Curve shifts: from AB to A'C

- Now, price inflation is 4%
- In the next wage round, wage setter will increase wages by 4% to cover the erosion of the expected real wage plus an additional 2% given the bargaining gap
- Nominal wages increase by 6% in total and so does Price inflation
- The new PC curve ($A'C$) is higher than the old one (AB)
- Vice versa, a negative demand shock which generates a negative bargaining gap will result in lower employment and disinflation

Modelling: Summary of the PC

Wage setters have Adaptive inflation expectations: $\pi_t^E = \pi_{t-1}$

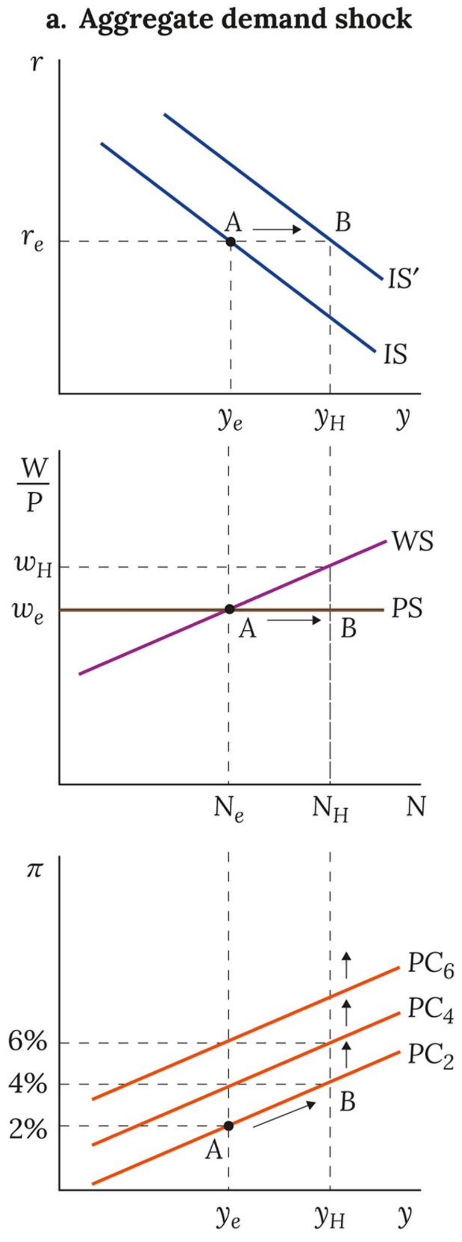
- They update their inflation expectations based on last period's level

PC is upward sloping: if $y_t > y_e$, then $\pi_t > \pi_{t-1}$

$$\pi_t = \pi_t^E + \alpha(y_t - y_e) \qquad \text{(adaptive expectations Phillips Curve)}$$

Adaptive Expectations PC is characterised by:

- Lagged inflation (π_{t-1}) fixes the PC's vertical height above y_e
- Slope of the WS curve determines the PC's slope



Application: Aggregate Demand Shock

Initial Equilibrium at A $(y_e, r_e, w_e, N_e; \pi_{t-1} = 2\%)$

1. Positive AD Shock → IS shifts right
2. No stabilizing policy → r_e is fixed
3. Output increases to y_H & positive bargaining gap opens
4. Wage setters try to set w_H & Price setters increase prices to restore profit margin
5. Price inflation increases to 4%
6. Same process next period, since the bargaining gap persists

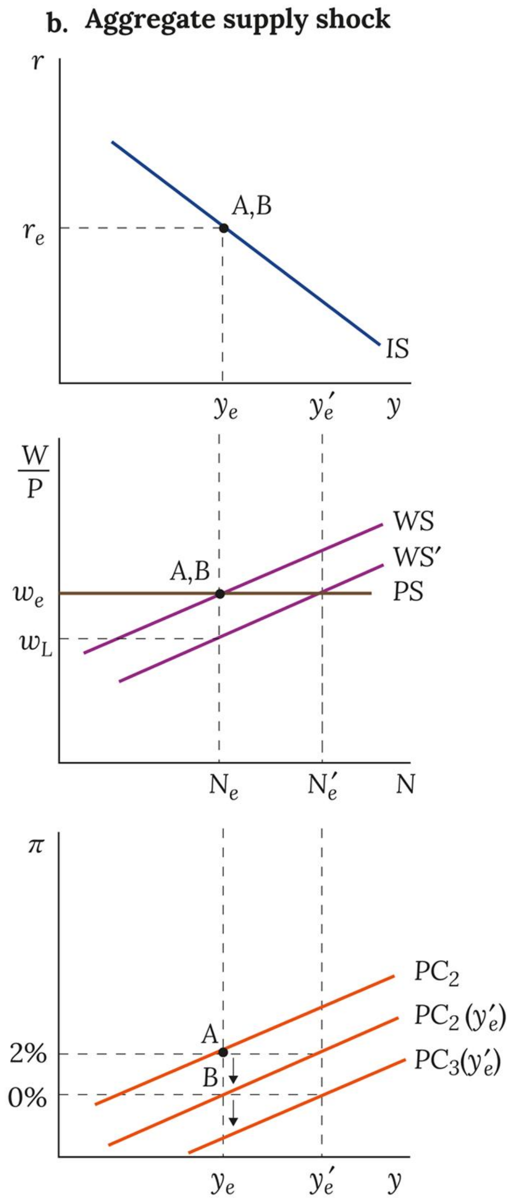
Hence, without stabilising policy (r_e constant), positive AD shock is accompanied by ever-increasing inflation

Figure 2.18(a) World without a stabilizing policy maker: inflationary implications of aggregate demand and supply-side shocks. Visit www.oup.com/he/carlin-soskice to explore shocks without a stabilizing policy maker with Animated Analytical Diagram 2.18.
Note: PC_x is used as short form for $PC(\pi^E_t = x\%)$, where x is the level of lagged inflation and t is the time period.

Application: Supply-side Shock

For supply-side shocks, there are more ways to implement in our model:

- Production function shifts or productivity shocks: changes in λ
- PS curve shifts: changes in competition condition in the product market (change in mark-up μ) or price-push factors ($\boldsymbol{z}_P$)
- WS curve shifts: changes in wage-push factors ($\boldsymbol{z}_w$) or shifts in bargaining power from workers to employers



Application: Supply-side Shock

Initial Equilibrium at A $(y_e, r_e, w_e, N_e, \pi_{t-1} = 2\%)$

1. Reduced unemployment benefits → WS shifts down
2. Equilibrium y_e increases to y_e' and N_e increases to N_e'
3. No stabilizing policy → r_e is fixed
4. Negative bargaining gap (-2%) at y_e
5. Inflation falls by 2% → nominal wages don't change
6. Inflation falls every period as long as y < $y_e{}'$

Hence, without stabilising policy (r_e constant), positive supply shock is associated with falling inflation at the initial output level $y = y_e$

Figure 2.18(a) World without a stabilizing policy maker: inflationary implications of aggregate demand and supply-side shocks. Visit www.oup.com/he/carlin-soskice to explore shocks without a stabilizing policy maker with Animated Analytical Diagram 2.18.
Note: PC_x is used as short form for $PC(\pi^E_t = x\%)$, where x is the level of lagged inflation and t is the time period.

Summary

The intersection of the WS and PS curves pins down the equilibrium unemployment

- Equilibrium rate of unemployment changes with:
  - structural features of the economy
  - policy choices of unemployment benefits
  - the evolution of market power
  - stringency of competition policy

If output is off-equilibrium, inflation moves away from its medium-run equilibrium level reflecting the conflicting claims on output per worker of workers and owners

Summary

In the absence of a policy maker:

- A positive demand shock leads to a continuously increasing inflation rate as long as there is a positive output gap
- A positive supply shock leads to an increase in equilibrium output and is associated with falling inflation as long as there is a negative output gap

Shifts in the WS-PS model can be translated into changes in the Lorenz curve and hence reflect impact on inequality, measured by the Gini coefficient